# Distributions and critical values

This sheet is a compact lookup for the probability models used in S3 and the sampling and inference methods in S6–S9. A random variable is written with an upper-case letter (for example, $X$); $x$ denotes a realised value. A PMF is written $p_X(x)$, a PDF $f_X(x)$, and a CDF is $F_X(x) := \mathbb{P}(X \leq x)$. Values outside a stated support have PMF or PDF equal to zero.

## How to use the table

1. Identify whether the variable is discrete or continuous, then check the parameterisation, restrictions and support before substituting numbers.
2. For a discrete variable, sum the PMF over the required support points. For a continuous variable, integrate the PDF over an interval; a single point has probability zero. Use the CDF for intervals and tails.
3. A $q$-quantile is
$$x_q := \inf\{x : F_X(x) \geq q\}, \qquad 0 < q < 1.$$
For continuous distributions, $F_X(x_q) = q$. If $Z \sim \mathcal{N}(0,1)$, write
$$\Phi(z) := \mathbb{P}(Z \leq z), \qquad z_q := \Phi^{-1}(q).$$
If $T \sim t_\nu$, write
$$t_{q,\nu} := \inf\{t : \mathbb{P}(T \leq t) \geq q\}.$$
Both reference distributions are symmetric, so $z_{1-q} = -z_q$ and $t_{1-q,\nu} = -t_{q,\nu}$. The tables give positive upper quantiles. A lower-tail critical value is its negative; a two-sided level-$\alpha$ procedure uses the $1 - \alpha/2$ column.

## Required probability distributions (S3)

The discrete distributions below use the PMF $p_X(x)$. Their means and variances are finite for every parameter value allowed in the table.

| Distribution (complete restrictions) | Support and PMF | $\mathbb{E}[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|
| $X \sim$ Bernoulli$(p)$, $0 \leq p \leq 1$. One trial; $X = 1$ denotes success. | $\{0,1\}$; $p_X(1) = p$, $p_X(0) = 1 - p$. | $p$ | $p(1-p)$ |
| $X \sim$ Binomial$(n,p)$, $n \in \{1,2,...\}$, $0 \leq p \leq 1$. Number of successes in $n$ independent Bernoulli$(p)$ trials. | $\{0,1,\dots,n\}$; $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$. | $np$ | $np(1-p)$ |
| $X \sim$ Geometric$(p)$, $0 < p \leq 1$. **Course convention:** number of trials up to and including the first success. | $\{1,2,...\}$; $p_X(k) = (1-p)^{k-1} p$. | $1/p$ | $(1-p)/p^2$ |
| $X \sim$ Poisson$(\lambda)$, $\lambda > 0$. The parameter is both mean and rate. | $\{0,1,2,...\}$; $p_X(k) = e^{-\lambda}\lambda^k/k!$. | $\lambda$ | $\lambda$ |

At the allowed probability endpoints, use the corresponding degenerate laws: the binomial variable equals 0 almost surely when $p = 0$ and equals $n$ almost surely when $p = 1$; the geometric variable equals 1 almost surely when $p = 1$. This avoids treating $0^0$ as an ordinary power.

For example, the geometric tail is $\mathbb{P}(X > m) = (1-p)^m$ for $m = 0, 1, \ldots$. A discrete CDF is the corresponding cumulative sum, $F_X(x) = \sum_{k \le x} p_X(k)$, over the support.

The continuous distributions below use the PDF $f_X(x)$. Their CDFs are included where a short closed form is useful.

| Distribution (complete restrictions) | Support and PDF | CDF or tail | $\mathbb{E}[X]$ | $\mathrm{Var}(X)$ |
|---|---|---|---|---|
| $X \sim$ Uniform$(a, b)$, $a, b \in \mathbb{R}$, $a < b$. | $[a, b]$; $f_X(x) = 1/(b-a)$. | $F_X(x) = 0$ for $x < a$; $(x-a)/(b-a)$ for $a \le x \le b$; 1 for $x > b$. | $(a+b)/2$ | $(b-a)^2/12$ |
| $X \sim$ Exponential$(\lambda)$, $\lambda > 0$. **Course convention:** $\lambda$ is the rate. | $[0, \infty)$; $f_X(x) = \lambda e^{-\lambda x}$ for $x \ge 0$. | For $x \ge 0$, $F_X(x) = 1 - e^{-\lambda x}$ and $\mathbb{P}(X > x) = e^{-\lambda x}$; for $x < 0$, these are 0 and 1, respectively. | $1/\lambda$ | $1/\lambda^2$ |

**Normal distribution**

For $X \sim \mathcal{N}(\mu, \sigma^2)$, let $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, so $\sigma := \sqrt{\sigma^2} > 0$. **The second parameter is the variance.** Its support is $\mathbb{R}$, its density is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right],$$

its CDF is $F_X(x) = \Phi((x-\mu)/\sigma)$, and $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$.

For a linear transformation $Y = cX + d$, where $c, d \in \mathbb{R}$, whenever the moments exist,

$$\mathbb{E}[Y] = c\mathbb{E}[X] + d, \qquad \mathrm{Var}(Y) = c^2 \,\mathrm{Var}(X).$$

## Reference distributions for sampling and inference (S6–S9)

These are not additional population models required in S3.

| Distribution (complete restrictions) | Support and density | Mean | Variance |
|---|---|---|---|
| $T \sim t_\nu$, $\nu \in \{1, 2, \ldots\}$ degrees of freedom. | $\mathbb{R}$; $f_T(t) = \dfrac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + t^2/\nu\right)^{-(\nu+1)/2}$. | 0 if $\nu > 1$; undefined for $\nu = 1$. | $\nu/(\nu-2)$ if $\nu > 2$; infinite for $\nu = 2$, undefined for $\nu = 1$. |

| Distribution (complete restrictions) | Support and density | Mean | Variance |
|---|---|---|---|
| $Q \sim \chi^2_\nu$, $\nu \in \{1, 2, ...\}$ degrees of freedom. | $[0, \infty)$; for $q > 0$, $f_Q(q) = \frac{q^{\nu/2-1}e^{-q/2}}{2^{\nu/2}\Gamma(\nu/2)}$. The value assigned at the single point $q = 0$ does not affect the distribution. | $\nu$ | $2\nu$ |

Here $\Gamma$ denotes the gamma function. The $t_\nu$ density has finite mean only when $\nu > 1$, and finite variance only when $\nu > 2$, as shown in the table.

If $X_1, \dots, X_n$ are a normal random sample with $n \geq 2$, then

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}, \qquad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

where

$$\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Both results require a normal population. The first pivot is used for the course's Student's $t$ procedures when $\sigma^2$ is unknown; the identity itself does not require that the analyst be unaware of $\sigma^2$. Here $\chi^2_\nu$ is used only as a sampling/reference distribution; categorical chi-squared tests are outside this course.

## Normal and Student's $t$ critical values

The entries below are upper quantiles rounded to three decimal places. For a one-sided procedure with tail probability $\alpha$, use $q = 1 - \alpha$. For a two-sided $100(1-\alpha)\%$ interval or test, use $q = 1 - \alpha/2$. Thus the columns cover one-sided 90%, 95% and 99% procedures ($q = .90, .95, .99$) and two-sided 90%, 95% and 99% procedures ($q = .95, .975, .995$).

**Standard normal**

| $q$ | $z_q$ | Typical use |
|---|---|---|
| 0.900 | 1.282 | one-sided 90% |
| 0.950 | 1.645 | one-sided 95% or two-sided 90% |
| 0.975 | 1.960 | two-sided 95% |
| 0.990 | 2.326 | one-sided 99% |
| 0.995 | 2.576 | two-sided 99% |

**Student's $t$**

The rows cover the degrees of freedom used in the course's small-sample examples ($4, 8, 9, 15$); $\nu = \infty$ is the standard-normal limit.

| $\nu$ | $t_{.90,\nu}$ | $t_{.95,\nu}$ | $t_{.975,\nu}$ | $t_{.99,\nu}$ | $t_{.995,\nu}$ |
|---|---|---|---|---|---|
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 |

| $\nu$ | $t_{.90,\nu}$ | $t_{.95,\nu}$ | $t_{.975,\nu}$ | $t_{.99,\nu}$ | $t_{.995,\nu}$ |
|---|---|---|---|---|---|
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 |
| $\infty$ (normal limit) | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 |

## Sampling and inference reminders

### Samples, the LLN and the CLT (S6)

For an i.i.d. sample with $\mathbb{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$,

$$\mathbb{E}[\bar{X}] = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

The law of large numbers says $\bar{X} \xrightarrow{\mathbb{P}} \mu$ as $n \to \infty$ (under the stated i.i.d. finite-moment conditions). It is not a claim that observations become normal. If $\sigma^2 > 0$, the central limit theorem gives the large-$n$ approximation

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \;\approx\; \mathcal{N}(0,1).$$

When the population is normal, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly for every $n$; otherwise the CLT statement is generally approximate.

### Confidence intervals (S8)

For a normal population with known $\sigma$, an exact two-sided $100(1-\alpha)\%$ interval for $\mu$ is

$$\left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

For a normal population with unknown $\sigma^2$, use $S$ and the exact Student's $t$ pivot:

$$\left[\bar{X} - t_{1-\alpha/2,n-1}\frac{S}{\sqrt{n}}, \bar{X} + t_{1-\alpha/2,n-1}\frac{S}{\sqrt{n}}\right].$$

For an iid Bernoulli sample $X_1, \ldots, X_n$, let $K := \sum_{i=1}^n X_i \sim \mathrm{Binomial}(n,p)$ be the success count and let $\hat{p} := K/n = \bar{X}$. When $n\hat{p}$ and $n(1-\hat{p})$ are sufficiently large, the usual large-sample interval is

$$\hat{p} \;\pm\; z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

The confidence level describes repeated-sampling coverage when the method's conditions hold. After data are observed, the realised endpoints are fixed; do not describe them as assigning probability to a fixed parameter.

### Tests and errors (S9)

For $H_0 : \mu = \mu_0$, use

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \quad \text{(known variance)}, \qquad t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad \text{(normal population, unknown variance)}.$$

For iid observations, the known-variance $z$ statistic is exactly standard normal under $H_0$ when the population is normal and is approximately standard normal under the finite-variance, large-sample CLT conditions stated above. The $t$ result is exact under the displayed normal-population condition.

For an iid Bernoulli sample and $H_0 : p = p_0$, where $0 < p_0 < 1$, a large-sample proportion test uses

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}.$$

The normal approximation requires both null expected counts $np_0$ and $n(1-p_0)$ to be sufficiently large.

Choose the tail from $H_1$: reject in an upper tail when the statistic is large, in a lower tail when it is small, and in both tails when $H_1$ is two-sided. The $p$-value is the probability under $H_0$ of a result at least as incompatible with $H_0$ as the observed statistic; it is not the probability that $H_0$ is true. A Type I error has probability $\alpha$; at a specified alternative $\theta$, Type II error is $\beta(\theta)$ and power is $1 - \beta(\theta)$.

With the same standard error and normal or $t$ method, a two-sided level- $\alpha$ test of $H_0 : \mu = \mu_0$ rejects exactly when $\mu_0$ lies outside the corresponding $100(1-\alpha)\%$ confidence interval.

## Common pitfalls

- Do not replace the course geometric variable (trials **including** the success) by a failures-before-success convention.
- Do not treat an exponential parameter $\lambda$ as a mean, or the second normal parameter $\sigma^2$ as a standard deviation.
- A density value is not a probability; integrate a PDF over an interval.
- Check the $t$ degrees of freedom $(n-1)$ and the normal-population condition before using an exact small-sample $t$ interval or test.
- The LLN does not assert normality, and a $p$-value is not $\mathbb{P}(H_0 \text{ is true})$. Use $S^2$ with denominator $n-1$.